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# Mathematical Modelling of the Three-Dimensional Free Vibration of Bidirectionally Graded Porous Plates on a Kerr Foundation: A Rayleigh–Ritz Eigenvalue Approach

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**Abstract:** This paper develops a mathematical model for the three-dimensional (3D) free vibration of porous plates whose material properties are graded along the length and through the thickness and which rest on a three-parameter Kerr foundation. The problem is formulated as an eigenvalue problem for the 3D elasticity operator with coefficients that vary in two coordinates, and it is solved by the Rayleigh–Ritz method. The trial functions are numerically orthonormalised Legendre polynomials, the separable structure of the coefficients turns every matrix block into a Kronecker product, and the massless foundation field is eliminated by static condensation, which leads to a symmetric generalized matrix eigenvalue problem. The model is verified against an independent exact solution, obtained for thickness-dependent coefficients from a two-point boundary-value problem for six first-order differential equations, and against classical results for homogeneous plates; its convergence is examined for different edge conditions. Two reduced two-dimensional models, a higher-order shear deformation theory and a reduced-unknown quasi-3D theory, are formulated in the same framework and assessed against the 3D model. The results show that in-plane grading lowers the eigenvalues considerably and makes the eigenfunctions asymmetric, that a stiff foundation can turn an in-plane mode into the fundamental mode, and that porosity lowers the frequency of an unsupported plate but raises it on a stiff foundation. The higher-order theory agrees closely with the 3D model for simply supported plates of moderate and small thickness, whereas the quasi-3D theory retains a noticeable error even for very thin thickness-graded plates because of thickness locking. New 3D benchmark eigenvalues are provided.

**Keywords:** Mathematical modelling; Rayleigh–Ritz method; Generalized eigenvalue problem; State-space method; Functionally graded plates; Free vibrations; Kerr foundation

## 1 Introduction

Many problems of applied mechanics lead to eigenvalue problems for partial differential operators with variable coefficients. The free vibration of a plate whose material properties change from point to point is a typical example: the natural frequencies and mode shapes are the eigenvalues and eigenfunctions of the three-dimensional (3D) elasticity operator, whose coefficients follow the spatial distribution of the stiffness and the density. Closed-form solutions exist only when the coefficients depend on a single coordinate and the boundary conditions allow a separation of variables. In all other cases the problem must be approximated, and two questions then arise: how to construct a discretisation that is accurate, well conditioned and verifiable, and how much accuracy is lost when the 3D problem is replaced by a simpler two-dimensional (2D) model.

Functionally graded materials (FGMs) give rise to exactly such problems. Their composition changes continuously, typically from a ceramic on one face to a metal on the other [1]. When the service conditions also vary along a component, the composition can be graded in a second direction, which gives more freedom to tailor the stiffness and the thermal stresses than one-directional grading [2, 3]. The static and dynamic behaviour of these bidirectional FGMs has been studied for beams [4, 5] and plates [6–11]. Mathematically, grading along the length as well as through the thickness makes the coefficients of the governing equations depend on two coordinates. The Fourier-series solutions available for thickness-graded plates then no longer apply, even for simply supported edges.

For plates graded through the thickness only, the free-vibration problem is well understood. Vel and Batra [12] obtained exact 3D solutions for simply supported plates, Uymaz and Aydogdu [13] used a 3D Ritz method for other edge conditions, and Matsunaga [14] and Hosseini-Hashemi et al. [15] gave higher-order and first-order plate solutions. The Rayleigh–Ritz method with polynomial trial functions is particularly suited to such problems [13, 16, 17]: it handles arbitrary combinations of edge conditions, it converges from above to the exact eigenvalues, and the trial functions need to satisfy only the essential boundary conditions. Its main numerical difficulty is the ill-conditioning of the mass and stiffness matrices when high-degree polynomials are used, which limits the attainable accuracy. Reduced 2D models are an alternative. Shear deformation theories and quasi-3D theories, which add a transverse normal strain, have been proposed with full expansions [18, 19] and with a reduced number of unknowns [20–24]. They replace the 3D eigenvalue problem by a smaller one, but their accuracy has to be established against 3D solutions. Plates on two- and three-parameter elastic foundations have been analysed with such models [25–27], and Zhou et al. [16] gave 3D Ritz solutions for thick homogeneous plates on a Pasternak foundation.

For bidirectionally graded plates the available results are based on reduced models only. Melaibari et al. [7] studied the free vibration of bidirectional graded porous plates on elastic foundations with a 2D differential quadrature method, and Lieu et al. [11] analysed the bending and free vibration of in-plane bidirectional plates by isogeometric analysis. Kerr-foundation studies have addressed the thermal vibration of bidirectionally stepped porous plates [28] and the free vibration of thickness-graded porous plates with a quasi-3D theory [29]. To the author's knowledge, no 3D solutions are available for the free vibration of plates graded along the length and through the thickness on a Kerr foundation. Without them the accuracy of the reduced models cannot be assessed. This matters because, for the same plates under mechanical load, the reduced-unknown quasi-3D theory was found to be slightly stiffer than the 3D solution because of thickness locking, and in-plane grading shifted the peak deflection away from the plate centre [30]. Both effects can be expected to influence the eigenvalues and eigenfunctions as well.

This paper develops and verifies a mathematical model for this eigenvalue problem. The main contributions are as follows:

1. A 3D Rayleigh–Ritz formulation is constructed for a plate whose stiffness and density vary in two coordinates and which rests on a three-parameter Kerr foundation. The trial functions are Legendre polynomials with boundary factors, orthonormalised numerically to keep the matrices well conditioned. The separable structure of the coefficients is exploited so that every matrix block becomes a Kronecker product. The massless foundation field is eliminated by static condensation, which leads to a symmetric generalized eigenvalue problem for the plate alone.
2. An independent exact solution is derived for the case in which the coefficients depend on the thickness coordinate only. The problem is reduced to a two-point boundary-value problem for a system of six first-order ordinary differential equations, and the eigenvalues are obtained as roots of a boundary determinant. This solution, together with classical results for homogeneous plates, verifies the Ritz model, and its convergence is examined for different edge conditions.
3. Two reduced 2D models, a higher-order shear deformation theory (HSDT) and a reduced-unknown quasi-3D theory, are formulated within the same Rayleigh–Ritz framework and assessed against the 3D model over a range of thickness ratios, gradation indices and edge conditions.
4. New 3D benchmark eigenvalues are provided, and a parametric study shows how the gradation indices, porosity, the three foundation parameters, the edge conditions and the thickness ratio affect the eigenvalues and the shape of the eigenfunctions. A kinetic-energy criterion is introduced to distinguish flexural from in-plane modes.

The paper is organised as follows. Section 2 presents the mathematical model, Section 3 the Rayleigh–Ritz discretisation, Section 4 the verification, the assessment of the reduced models and the parametric study, and Section 5 the conclusions.

## 2 Mathematical Model

Consider a rectangular plate of length $a$, width $b$ and thickness $h$, occupying $0 \le x \le a$, $0 \le y \le b$, $-h/2 \le z \le h/2$, which rests on a three-parameter Kerr foundation (Figure 1). The plate and the foundation follow the model developed in a previous study [30], and only the ingredients needed for free vibration are summarised here.

### 2.1 Bidirectional Porous Material

The ceramic volume fraction changes along the length and through the thickness,

$$V_c(x,z) = \left(\frac{x}{a}\right)^{k_x} \left(\frac{1}{2} + \frac{z}{h}\right)^{k_z}, \tag{1}$$

so that the bottom face and the edge $x = 0$ are pure metal, while the top face becomes pure ceramic at $x = a$. With an even porosity of volume fraction $\alpha_p$, the modified rule of mixtures [31] gives the Young modulus and the mass density as

$$E = E_m + (E_c - E_m)V_c - \frac{\alpha_p}{2}(E_c + E_m),$$
$$\rho = \rho_m + (\rho_c - \rho_m)V_c - \frac{\alpha_p}{2}(\rho_c + \rho_m). \tag{2}$$

Poisson's ratio is constant, $\nu = 0.3$. The density follows the same rule as the modulus, as is customary for porous FG plates [7, 31]. Porosity therefore lowers both the stiffness and the inertia of the plate, but in different proportions: at the metal-rich bottom face it removes $\alpha_p(E_c + E_m)/(2E_m) = 3.2\alpha_p$ of the modulus and only $\alpha_p(\rho_c + \rho_m)/(2\rho_m) = 1.2\alpha_p$ of the density.

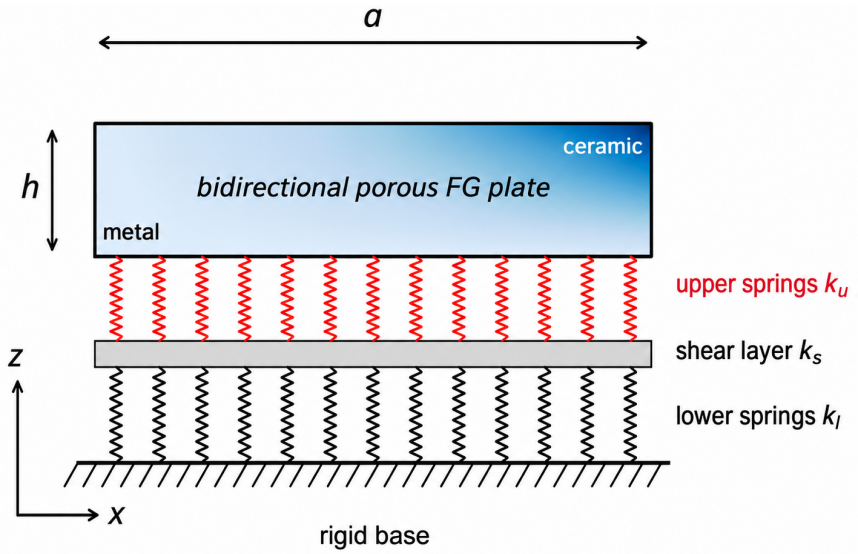



**Figure 1.** Bidirectional porous functionally graded (FG) plate on a three-parameter Kerr foundation (side view)
Note: FG = functionally graded; $a$ = plate length; $h$ = plate thickness; $k_u$ = stiffness of the upper spring layer; $k_s$ = stiffness of the shear layer; $k_l$ = stiffness of the lower spring layer; the shading indicates the ceramic volume fraction.

### 2.2 Kerr Foundation

The Kerr foundation [32] comprises an upper spring layer of stiffness $k_u$, a shear layer of stiffness $k_s$ and a lower spring layer of stiffness $k_l$. The deflection of the shear layer, $w_f(x, y)$, is kept as an independent field, and the foundation stores the energy:

$$U_f = \frac{1}{2}\int_0^a \int_0^b \left[k_u(w_0 - w_f)^2 + k_l w_f^2 + k_s(w_{f,x}^2 + w_{f,y}^2)\right] dx\, dy, \tag{3}$$

where, $w_0 = w(x, y, -h/2)$ is the deflection of the bottom face. The foundation is massless, so $w_f$ carries no inertia. The Pasternak model [33] is recovered as $k_u \to \infty$, and the Winkler model [34] when, in addition, $k_s = 0$. The foundation parameters are written in the dimensionless form $K_u = k_u a^4/D_c$, $K_l = k_l a^4/D_c$, and $K_s = k_s a^2/D_c$, with $D_c = E_c h^3/[12(1 - \nu^2)]$.

### 2.3 Energy Functional

For harmonic motion with circular frequency $\omega$, the displacements are $\{u, v, w\}e^{i\omega t}$ and $w_f e^{i\omega t}$. The free vibrations make the Rayleigh–Ritz functional

$$\Pi = \frac{1}{2}\int_V \varepsilon^T C \varepsilon \, dV + U_f - \frac{\omega^2}{2}\int_V \rho(u^2 + v^2 + w^2)\, dV \tag{4}$$

stationary, where $C(x, z) = E(x, z)\hat{C}(\nu)$ is the isotropic stiffness at each point. Eq. (4) contains no assumption about the through-thickness kinematics; the three models below differ only in how $u$, $v$ and $w$ are approximated.

### 2.4 Three-Dimensional Elasticity Model

In the 3D model each displacement component is expanded in all three coordinates,

$$\{u, v, w\} = \sum_{i,j,k} \{a_{ijk}, b_{ijk}, c_{ijk}\} X_i(\xi) Y_j(\eta) P_k(2\zeta - 1), \tag{5}$$

with $\xi = x/a$, $\eta = y/b$, $\zeta = z/h + 1/2$, and $P_k$ the Legendre polynomial of degree $k$. The full 3D stiffness is used and the transverse normal and shear strains are retained, so the model is exact apart from the truncation of the series. Three-dimensional Ritz solutions with polynomial bases are well established for the vibration of thick homogeneous and FG plates [13, 16, 17]. The foundation acts on $w(x, y, -h/2)$. A simply supported edge is "hard", i.e. the tangential and transverse displacements vanish over the whole edge face.

### 2.5 Plate Models

The HSDT of Zenkour [35] and the reduced-unknown quasi-3D sinusoidal theory of Thai and Kim [20] are adopted in this study. The displacement field is

$$\begin{aligned} u &= u_0 - z w_{b,x} - f(z) w_{s,x}, \\ v &= v_0 - z w_{b,y} - f(z) w_{s,y}, \\ w &= w_b + w_s + g(z) w_z, \end{aligned} \tag{6}$$

with $f(z) = z - (h/\pi)\sin(\pi z/h)$ and $g(z) = 1 - f'(z) = \cos(\pi z/h)$. In the HSDT $w_z = 0$ and plane stress is assumed; the quasi-3D theory keeps $w_z$, the transverse normal strain $\varepsilon_z = g'(z) w_z$ and the full 3D stiffness. The kinetic energy is evaluated from Eq. (6) with all inertia terms, including the rotary and higher-order ones. Because $E$ and $\rho$ vary with $x$, the stiffness and inertia resultants are functions of $x$ and are integrated numerically.

## 3 Rayleigh–Ritz Discretisation

The in-plane functions of all fields, including the shear-layer deflection $w_f$, are

$$X_i(\xi) = \xi^{p_0} (1 - \xi)^{p_1} P_i(2\xi - 1), \quad i = 0, \ldots, N - 1, \tag{7}$$

and similarly $Y_j(\eta)$. The exponents $p_0$ and $p_1$ impose only the essential conditions at each edge [30]. On an edge normal to $x$, $v = w = 0$ for a simply supported (S) edge and $u = v = w = 0$ for a clamped (C) edge (with, in the plate models, $w_{b,\,x} = w_{s,\,x} = 0$); a free (F) edge has no constraint. The shear-layer deflection vanishes on S and C edges. Edges are named in the order $x = 0$, $y = 0$, $x = a$, $y = b$. To keep the algebraic system well conditioned for large $N$, each set of one-dimensional (1D) functions is orthonormalised numerically before use; this changes the basis but not the space it spans.

Substituting the expansions into Eq. (4) and making $\Pi$ stationary gives

$$\begin{bmatrix} K_{pp} & K_{pf} \\ K_{fp} & K_{ff} \end{bmatrix} \begin{bmatrix} c \\ e \end{bmatrix} = \omega^2 \begin{bmatrix} M & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c \\ e \end{bmatrix}, \tag{8}$$

where, $c$ collects the plate coefficients and $e$ those of $w_f$. Since the foundation is massless, $e$ is condensed statically, which leaves the symmetric eigenvalue problem

$$(K_{pp} - K_{pf} K_{ff}^{-1} K_{fp}) c = \omega^2 M c. \tag{9}$$

Because $E$ and $\rho$ depend on $x$ and $z$ only, every integral factorises into an $x$–$z$ part and a $y$ part, and each block of the matrices is a Kronecker product. The $x$–$z$ integrals are evaluated by Gauss–Legendre quadrature, with additional points for fractional $k_x$, for which the power law is not a polynomial. The lowest eigenvalues are obtained by shift-invert Lanczos iteration for the 3D model and by a dense solver for the plate models.

To separate flexural from in-plane modes, the fraction of the kinetic energy carried by $w$ is computed for every mode, $\chi = \dfrac{\int \rho w^2 \, dV}{\int \rho (u^2 + v^2 + w^2) \, dV}$. A mode is called flexural if $\chi > 0.5$ and in-plane otherwise. For plates graded only through the thickness the in-plane modes have $\chi = 0$. In bidirectional plates, bending and stretching are coupled along the plate, so the modes are mixed to some degree. For $a/h \geq 10$ the flexural modes computed here have $\chi > 0.8$ and the in-plane modes $\chi < 0.11$. Strongly mixed modes ($0.25 \leq \chi \leq 0.8$) occur only among the higher modes of thick plates ($a/h = 5$), where flexural and in-plane frequencies are close.

## 4 Numerical Results

The plates are made of alumina (ceramic) and aluminium (metal), with the properties in Table 1 and $\nu = 0.3$. The natural frequencies are reported as

$$\Omega = \omega a^2 \sqrt{\frac{\rho_c h}{D_c}}, \tag{10}$$

so that $\Omega = 2\pi^2 = 19.74$ for the fundamental mode of a thin, simply supported square ceramic plate without a foundation. Unless stated otherwise, the plate is square with $a/h = 10$, $k_x = 2$, $k_z = 1$, $\alpha_p = 0.1$, $(K_u, K_s, K_l) = (100, 10, 100)$ and SSSS edges. The 3D model uses $N = 14$ in-plane terms for SSSS plates and $N = 16$ when an edge is clamped, with $N_z = 10$ through the thickness; the plate models use $N = 16$ ($N = 20$ with clamped edges). In the tables, in-plane modes ($\chi < 0.5$) are marked with an asterisk.

**Table 1.** Material properties of the constituents

| Constituent | $E$ (GPa) | $\rho$ (kg/m$^3$) | $\nu$ |
|---|---|---|---|
| Alumina, $Al_2O_3$ (ceramic) | 380 | 3,800 | 0.3 |
| Aluminium, Al (metal) | 70 | 2,707 | 0.3 |

Note: $E$ = Young modulus; $\rho$ = mass density; $\nu$ = Poisson's ratio.

### 4.1 Validation and Convergence

**Homogeneous plate.** For a homogeneous, simply supported square plate with $a/h = 10$, Table 2 compares the frequency parameter $\lambda = \omega h\sqrt{\rho/G}$ with the classical exact 3D values of Srinivas et al. [36]. The independent state-space solution described below and the 3D Ritz model agree with each other to at least nine digits and with the published values to all reported digits (0.09315 rounds to 0.0932 [36]).

**Table 2.** Homogeneous SSSS plate, $a/h = 10$, $\nu = 0.3$: $\lambda = \omega h\sqrt{\rho/G}$ for modes $(m, n)$

| Solution | (1,1) | (1,2) | (2,2) |
|---|---|---|---|
| Srinivas et al. [36] | 0.09320 | 0.22260 | 0.34210 |
| Present, exact state-space | 0.09315 | 0.22260 | 0.34207 |
| Present, 3D Ritz | 0.09315 | 0.22260 | 0.34207 |

Note: 3D = three-dimensional; SSSS = all four edges simply supported; $a/h$ = length-to-thickness ratio; $\nu$ = Poisson's ratio; $\lambda$ = dimensionless frequency parameter; $\omega$ = circular frequency; $\rho$ = mass density; $G$ = shear modulus; $m$ and $n$ = mode numbers in the $x$ and $y$ directions.

**Graded plates on a Kerr foundation.** For $k_x = 0$ and hard simply supported edges, the 3D problem has an exact solution. With $u = U(z)\cos(m\pi x/a)\sin(n\pi y/b)$, $v = V(z)\sin(m\pi x/a)\cos(n\pi y/b)$ and $w = W(z)\sin(m\pi x/a)\sin(n\pi y/b)$, the equations of motion reduce to six first-order ordinary differential equations (ODEs) for the state vector $\{U, V, W, \tau_{xz}, \tau_{yz}, \sigma_z\}$. The shear stresses vanish on both faces, $\sigma_z = 0$ on the top face, and the Kerr foundation gives $\sigma_z = k_{\text{eq}} W$ on the bottom face, with $k_{\text{eq}} = k_u(k_l + k_s\lambda_{mn}^2)/(k_u + k_l + k_s\lambda_{mn}^2)$ and $\lambda_{mn}^2 = (m\pi/a)^2 + (n\pi/b)^2$. The system was integrated with an eighth-order Runge–Kutta method at a relative tolerance of $10^{-12}$, and the frequencies were found as roots of the $3 \times 3$ boundary determinant. This solution shares no code with the Ritz model. Table 3 shows that the 3D Ritz model reproduces it to all reported digits for two thickness ratios, two thickness indices and three foundations.

**Convergence.** Table 4 lists the first six frequencies of the reference bidirectional plate for increasing $N$ and $N_z$. For the SSSS plate all six frequencies are converged to the reported digits from $N = 10$. With clamped edges the frequencies decrease monotonically and slowly with $N$, because the 3D clamped condition prescribes all three displacements over the whole edge face. Between $N = 16$ and $N = 18$ the clamped frequencies still decrease by up to about 0.01, and the successive differences decay approximately as $N^{-2.3}$, which reflects the singular behaviour of the 3D field at the clamped edges. An extrapolation of the sequence $N = 14, 16, 18$ to $N \to \infty$ (rows with $N = \infty$ in Table 4) shows that the clamped values obtained with $N = 16$ exceed the converged values by at most about 0.04. Since Rayleigh–Ritz eigenvalues converge from above, the clamped results are upper bounds that are accurate to about three significant digits, whereas the SSSS results are accurate to all reported digits. The through-thickness expansion $N_z = 10$ is sufficient in all cases: changing $N_z$ from 8 to 12 alters the frequencies by at most one unit in the fourth decimal.

### 4.2 Accuracy of the Plate Theories

Table 5 compares the first two flexural frequencies from the two plate theories with the 3D solution. The quasi-3D theory overestimates all the frequencies, and the HSDT overestimates the fundamental frequency in every case; for

the second flexural frequency of the thickness-graded SSSS plates ($k_x = 0$), however, the HSDT gives slightly lower values than the 3D solution. The two errors also behave differently as the plate becomes thinner.

**Table 3.** SSSS, $k_x = 0$, $\alpha_p = 0.1$: present 3D Ritz model versus the exact state-space solution

| $a/h$ | $k_z$ | $(K_u, K_s, K_l)$ | $\Omega_{11}$ Present | $\Omega_{11}$ Exact | $\Omega_{12}$ Present | $\Omega_{12}$ Exact |
|---|---|---|---|---|---|---|
| 5 | 1 | – | 13.2795 | 13.2795 | 29.5920 | 29.5920 |
| 5 | 1 | (100, 10, 100) | 16.1150 | 16.1150 | 30.9719 | 30.9719 |
| 5 | 1 | (1000, 10, 100) | 20.6036 | 20.6036 | 34.9315 | 34.9315 |
| 5 | 5 | – | 10.3850 | 10.3850 | 22.4119 | 22.4119 |
| 5 | 5 | (100, 10, 100) | 14.1586 | 14.1586 | 24.4207 | 24.4207 |
| 5 | 5 | (1000, 10, 100) | 19.3512 | 19.3512 | 29.4386 | 29.4386 |
| 10 | 1 | – | 14.3071 | 14.3071 | 34.3925 | 34.3925 |
| 10 | 1 | (100, 10, 100) | 17.2579 | 17.2579 | 35.8308 | 35.8308 |
| 10 | 1 | (1000, 10, 100) | 22.1246 | 22.1246 | 40.2613 | 40.2613 |
| 10 | 5 | – | 11.5117 | 11.5117 | 27.2355 | 27.2355 |
| 10 | 5 | (100, 10, 100) | 15.4300 | 15.4300 | 29.2530 | 29.2530 |
| 10 | 5 | (1000, 10, 100) | 21.3234 | 21.3234 | 35.1320 | 35.1320 |

Note: 3D = three-dimensional; SSSS = all four edges simply supported; $a/h$ = length-to-thickness ratio; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; $\alpha_p$ = porosity volume fraction; $\Omega_{11}$ and $\Omega_{12}$ = the dimensionless frequencies of the (1,1) and (1,2) flexural modes, where $\Omega = \omega a^2(\rho_c h/D_c)^{1/2}$ is the dimensionless frequency; Present = the 3D Rayleigh–Ritz model; Exact = the independent state-space solution; $(K_u, K_s, K_l)$ = the dimensionless Kerr parameters; "–" = a plate without a foundation. All plates are simply supported on all four edges (SSSS), with $k_x = 0$ and $\alpha_p = 0.1$.

**Table 4.** Convergence of the 3D model; $k_x = 2$, $k_z = 1$, $\alpha_p = 0.1$, $a/h = 10$, Kerr (100, 10, 100)

| BC | $N$ | $N_z$ | $\Omega_1$ | $\Omega_2$ | $\Omega_3$ | $\Omega_4$ | $\Omega_5$ | $\Omega_6$ |
|---|---|---|---|---|---|---|---|---|
| SSSS | 10 | 10 | 15.0498 | 27.7255 | 27.9721 | 38.4081* | 38.8950* | 41.3195 |
| SSSS | 12 | 10 | 15.0498 | 27.7255 | 27.9721 | 38.4081* | 38.8950* | 41.3194 |
| SSSS | 14 | 10 | 15.0498 | 27.7255 | 27.9721 | 38.4081* | 38.8950* | 41.3194 |
| SSSS | 16 | 10 | 15.0498 | 27.7255 | 27.9721 | 38.4081* | 38.8950* | 41.3194 |
| SSSS | 14 | 8 | 15.0498 | 27.7255 | 27.9721 | 38.4081* | 38.8950* | 41.3194 |
| SSSS | 14 | 12 | 15.0498 | 27.7255 | 27.9721 | 38.4081* | 38.8950* | 41.3194 |
| CSCS | 10 | 10 | 18.5798 | 29.7890 | 35.7717 | 38.8950* | 46.5051 | 49.2569 |
| CSCS | 12 | 10 | 18.5718 | 29.7823 | 35.7511 | 38.8950* | 46.4871 | 49.2506 |
| CSCS | 14 | 10 | 18.5675 | 29.7787 | 35.7397 | 38.8950* | 46.4772 | 49.2471 |
| CSCS | 16 | 10 | 18.5650 | 29.7765 | 35.7326 | 38.8950* | 46.4711 | 49.2450 |
| CSCS | 18 | 10 | 18.5633 | 29.7750 | 35.7279 | 38.8950* | 46.4670 | 49.2435 |
| CSCS | $\infty$ | 10 | 18.5585 | 29.7705 | 35.7129 | 38.8950* | 46.4537 | 49.2389 |
| CSCS | 14 | 8 | 18.5675 | 29.7787 | 35.7397 | 38.8950* | 46.4773 | 49.2471 |
| CSCS | 14 | 12 | 18.5675 | 29.7787 | 35.7397 | 38.8950* | 46.4772 | 49.2471 |
| CCCC | 10 | 10 | 21.3638 | 36.5628 | 37.3551 | 51.3161 | 57.6835 | 60.3496 |
| CCCC | 12 | 10 | 21.3499 | 36.5378 | 37.3304 | 51.2834 | 57.6450 | 60.3095 |
| CCCC | 14 | 10 | 21.3424 | 36.5238 | 37.3168 | 51.2654 | 57.6230 | 60.2877 |
| CCCC | 16 | 10 | 21.3380 | 36.5152 | 37.3084 | 51.2542 | 57.6091 | 60.2741 |
| CCCC | 18 | 10 | 21.3351 | 36.5095 | 37.3029 | 51.2468 | 57.6000 | 60.2651 |
| CCCC | $\infty$ | 10 | 21.3265 | 36.4910 | 37.2852 | 51.2227 | 57.5701 | 60.2363 |
| CCCC | 14 | 8 | 21.3424 | 36.5238 | 37.3168 | 51.2654 | 57.6231 | 60.2878 |
| CCCC | 14 | 12 | 21.3424 | 36.5238 | 37.3168 | 51.2654 | 57.6230 | 60.2877 |

Note: Edge conditions are given in the order $x = 0$, $y = 0$, $x = a$, $y = b$, with S = simply supported, C = clamped; 3D = three-dimensional; $a/h$ = length-to-thickness ratio; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; $\alpha_p$ = porosity volume fraction; BC = boundary conditions; $\Omega_1$–$\Omega_6$ = the first six dimensionless frequencies; SSSS = all four edges simply supported; CSCS = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ simply supported; CCCC = all four edges clamped; $N$ = the number of in-plane terms in each direction; $N_z$ = the number of through-thickness terms; "*" = marks in-plane modes (kinetic-energy fraction of the transverse displacement below 0.5). The rows with $N = \infty$ give the values extrapolated from the results for $N = 14$, 16 and 18 with $N_z = 10$.

The HSDT error is a pure thickness effect. For SSSS plates it is noticeable at $a/h = 5$, small at $a/h = 10$ and very small at $a/h = 20$. It is somewhat larger for bidirectional plates than for the thickness-graded plate at $a/h = 5$, and clearly larger for CSCS plates, for which it is the largest of all the cases considered at $a/h = 5$. Since the clamped 3D values converge from above (Table 4), these CSCS differences are, if anything, slightly underestimated.

The quasi-3D theory behaves differently. For the thickness-graded plate ($k_x = 0$, $k_z = 1$) its error does not vanish as the plate becomes thin. It remains practically unchanged from $a/h = 10$ to $a/h = 50$, whereas the HSDT error becomes negligible. This persistent error is the signature of thickness locking [37], identified for the same

plates under mechanical loading [30]. The single stretching function gives a transverse normal strain $g'(z)w_z$ that is odd in $z$. In a graded plate the bending strains are measured from a neutral surface that does not coincide with the mid-plane, so the Poisson contraction through the thickness, $\varepsilon_z \approx -\nu(\varepsilon_x + \varepsilon_y)/(1-\nu)$, has an even part that the theory cannot represent. The in-plane stiffness is then governed by the 3D coefficient $C_{11}$ instead of the plane-stress coefficient $Q_{11}$, which stiffens the plate. The effect is weaker in bidirectional plates, where the error of thin SSSS plates is small, because for $k_x \geq 2$ most of the plate is nearly metallic and therefore nearly homogeneous through its thickness. For thick plates the quasi-3D error can be smaller than that of the HSDT ($k_x = 5$, $a/h = 5$), but this is a cancellation of two errors of opposite sign rather than a sign of better kinematics.

For free vibration, therefore, both theories are acceptable for engineering purposes when $a/h \geq 10$ and the plate is simply supported. The HSDT is more consistent, because its error vanishes for thin plates. The reduced-unknown quasi-3D theory offers no advantage over the HSDT for these plates, and its error for thickness-graded plates should be kept in mind when it is used to generate reference values. The HSDT is used, together with 3D checks, in the parametric study below.

**Table 5.** First two flexural frequencies from the plate theories versus 3D elasticity; $\alpha_p = 0.1$, Kerr (100, 10, 100)

| BC | $a/h$ | $(k_x, k_z)$ | $\Omega_1$ 3D | $\Omega_1$ HSDT | $\Omega_1$ Quasi-3D | $\Omega_2$ 3D | $\Omega_2$ HSDT | $\Omega_2$ Quasi-3D |
|---|---|---|---|---|---|---|---|---|
| SSSS | 5 | (0, 1) | 16.1150 | 16.3288 | 16.4653 | 30.9719 | 30.9663 | 31.5353 |
| SSSS | 5 | (2, 1) | 14.0654 | 14.3664 | 14.2542 | 23.7705 | 23.9934 | 24.0208 |
| SSSS | 5 | (5, 1) | 13.4799 | 13.7791 | 13.6122 | 21.5586 | 21.8105 | 21.7210 |
| SSSS | 5 | (2, 5) | 13.4376 | 13.7647 | 13.6401 | 21.8094 | 22.0550 | 22.0739 |
| SSSS | 10 | (0, 1) | 17.2579 | 17.3085 | 17.5235 | 35.8308 | 35.7975 | 36.4700 |
| SSSS | 10 | (2, 1) | 15.0498 | 15.1208 | 15.1315 | 27.7255 | 27.7648 | 27.8970 |
| SSSS | 10 | (5, 1) | 14.3860 | 14.4532 | 14.4258 | 25.0514 | 25.0952 | 25.1274 |
| SSSS | 10 | (2, 5) | 14.4783 | 14.5521 | 14.5603 | 25.7822 | 25.8337 | 25.9592 |
| SSSS | 20 | (0, 1) | 17.5986 | 17.6111 | 17.8472 | 37.6128 | 37.6009 | 38.2961 |
| SSSS | 20 | (2, 1) | 15.3409 | 15.3586 | 15.4001 | 29.2278 | 29.2360 | 29.3825 |
| SSSS | 20 | (5, 1) | 14.6483 | 14.6648 | 14.6721 | 26.3863 | 26.3954 | 26.4462 |
| SSSS | 20 | (2, 5) | 14.7826 | 14.8006 | 14.8429 | 27.3259 | 27.3366 | 27.4896 |
| CSCS | 5 | (0, 1) | 19.7347 | 20.4575 | 20.8801 | 32.4963 | 33.1608 | 33.8961 |
| CSCS | 5 | (2, 1) | 16.3924 | 17.0910 | 17.1970 | 24.7690 | 25.5533 | 25.7393 |
| CSCS | 5 | (5, 1) | 15.6193 | 16.1757 | 16.1930 | 22.4517 | 23.0720 | 23.1070 |
| CSCS | 5 | (2, 5) | 15.3242 | 15.9787 | 16.0403 | 22.7019 | 23.3985 | 23.5097 |
| CSCS | 10 | (0, 1) | 22.4465 | 22.6937 | 23.1578 | 38.6817 | 39.0141 | 39.8489 |
| CSCS | 10 | (2, 1) | 18.5650 | 18.8143 | 18.9759 | 29.7765 | 30.1525 | 30.4080 |
| CSCS | 10 | (5, 1) | 17.5729 | 17.7554 | 17.8488 | 26.8278 | 27.1018 | 27.2259 |
| CSCS | 10 | (2, 5) | 17.4194 | 17.6556 | 17.8012 | 27.5091 | 27.8509 | 28.0781 |
| CSCS | 20 | (0, 1) | 23.4183 | 23.4645 | 23.9073 | 41.1945 | 41.2895 | 42.1181 |
| CSCS | 20 | (2, 1) | 19.3789 | 19.4354 | 19.5839 | 31.9096 | 32.0244 | 32.2561 |
| CSCS | 20 | (5, 1) | 18.3010 | 18.3403 | 18.4299 | 28.6934 | 28.7756 | 28.8855 |
| CSCS | 20 | (2, 5) | 18.2405 | 18.2967 | 18.4433 | 29.6582 | 29.7691 | 29.9964 |

Note: Edge conditions are given in the order $x = 0$, $y = 0$, $x = a$, $y = b$, with S = simply supported, C = clamped; 3D = three-dimensional; $a/h$ = length-to-thickness ratio; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; $\alpha_p$ = porosity volume fraction; BC = boundary conditions; $\Omega_1$ and $\Omega_2$ = the first two flexural dimensionless frequencies; HSDT = higher-order shear deformation; Quasi-3D = reduced-unknown quasi-3D theory theory; SSSS = all four edges simply supported; CSCS = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ simply supported.

### 4.3 Benchmark Results

Tables 6 and 7 give new 3D results for future comparison: the first six frequencies, with in-plane modes marked, for plates on the reference Kerr foundation (SSSS, CSCS and CCCC edges) and for SSSS plates without a foundation. Two porosities, two thickness indices and four length indices are covered. Based on Table 4, the SSSS values are accurate to all reported digits, and the values for plates with clamped edges to about three significant digits (as Rayleigh–Ritz values, they are upper bounds of the converged frequencies).

Several features of these results are worth noting. On the foundation, the in-plane modes of SSSS plates appear as the fourth and fifth modes, and those of CSCS plates as the fourth mode; they are almost unaffected by the foundation (compare Tables 6 and 7) and fall rapidly with $k_x$. In CCCC plates the in-plane modes lie above the first six flexural modes. For $k_x = 0$ the second and third modes of SSSS and CCCC plates are degenerate; for $k_x > 0$ they separate, and the separation grows with $k_x$. Porosity lowers the frequencies of the unsupported plates in all cases, whereas on the foundation its effect on the fundamental frequency of the thickness-graded SSSS plate is negligible (17.2541 vs. 17.2579).

### 4.4 Parametric Study

**Gradation indices.** Figure 2 shows the effect of the two indices on the fundamental frequency (HSDT lines, 3D symbols) and on the location $x_{\max}$ of the peak of the first mode along $y = b/2$. The frequency decreases

monotonically with both indices. In-plane grading replaces ceramic by aluminium, whose stiffness-to-density ratio $E/\rho$ is about four times lower than that of alumina, so the plate becomes softer faster than it becomes lighter. For $k_z = 1$ the 3D fundamental frequency falls from 17.26 at $k_x = 0$ to 15.05 at $k_x = 2$ and 14.39 at $k_x = 5$ on the foundation, a moderate reduction. Without a foundation the reduction is much larger, from 14.31 to 9.64, i.e. about one third, because the foundation stiffness does not depend on the composition. The steepest drop occurs for $k_x \leq 1$; beyond $k_x \approx 2$ the plate is mostly metallic and the frequency changes slowly.

**Table 6.** 3D benchmark frequencies; $a/h = 10$, Kerr (100, 10, 100)

| BC | $\alpha_p$ | $k_z$ | $k_x$ | $\Omega_1$ | $\Omega_2$ | $\Omega_3$ | $\Omega_4$ | $\Omega_5$ | $\Omega_6$ |
|---|---|---|---|---|---|---|---|---|---|
| SSSS | 0.0 | 1 | 0 | 17.2541 | 36.3790 | 36.3790 | 53.5156* | 53.5156* | 55.0008 |
| SSSS | 0.0 | 1 | 1 | 15.8434 | 31.6590 | 31.7062 | 43.6125* | 44.4200* | 47.3622 |
| SSSS | 0.0 | 1 | 2 | 15.2405 | 29.3863 | 29.5845 | 40.6709* | 41.3203* | 44.0543 |
| SSSS | 0.0 | 1 | 5 | 14.6266 | 27.0725 | 27.4772 | 36.5514* | 38.3674* | 40.5986 |
| SSSS | 0.0 | 5 | 0 | 15.7881 | 31.1649 | 31.1649 | 41.7303* | 41.7303* | 46.1020 |
| SSSS | 0.0 | 5 | 1 | 15.0994 | 28.9306 | 28.9821 | 37.3043* | 37.4859* | 42.6003 |
| SSSS | 0.0 | 5 | 2 | 14.7470 | 27.7491 | 27.8233 | 35.9262* | 36.3293* | 40.9178 |
| SSSS | 0.0 | 5 | 5 | 14.3715 | 26.4286 | 26.6473 | 34.3078* | 35.1538* | 39.0876 |
| SSSS | 0.1 | 1 | 0 | 17.2579 | 35.8308 | 35.8308 | 53.5088* | 53.5088* | 54.0641 |
| SSSS | 0.1 | 1 | 1 | 15.7143 | 30.4088 | 30.4588 | 41.7582* | 42.9276* | 45.2425 |
| SSSS | 0.1 | 1 | 2 | 15.0498 | 27.7255 | 27.9721 | 38.4081* | 38.8950* | 41.3194 |
| SSSS | 0.1 | 1 | 5 | 14.3860 | 25.0514 | 25.5310 | 33.3552* | 35.2034* | 37.2307 |
| SSSS | 0.1 | 5 | 0 | 15.4300 | 29.2530 | 29.2530 | 39.9613* | 39.9613* | 42.8949 |
| SSSS | 0.1 | 5 | 1 | 14.8238 | 27.0385 | 27.1091 | 34.4487* | 34.7441* | 39.3316 |
| SSSS | 0.1 | 5 | 2 | 14.4783 | 25.7822 | 25.8427 | 32.7793* | 33.1878* | 37.4846 |
| SSSS | 0.1 | 5 | 5 | 14.1020 | 24.3209 | 24.5566 | 30.7274* | 31.6721* | 35.4338 |
| CSCS | 0.0 | 1 | 0 | 22.6152 | 39.2733 | 47.3508 | 53.5156* | 62.5168 | 68.4222 |
| CSCS | 0.0 | 1 | 1 | 20.0292 | 34.0521 | 40.4298 | 43.6125* | 53.3090 | 58.3739 |
| CSCS | 0.0 | 1 | 2 | 19.2095 | 31.6105 | 38.0470 | 41.3203* | 49.6855 | 53.1440 |
| CSCS | 0.0 | 1 | 5 | 18.3353 | 29.0797 | 35.5537 | 38.3674* | 45.8613 | 48.1954 |
| CSCS | 0.0 | 5 | 0 | 19.8701 | 33.3264 | 39.4080 | 41.7303* | 51.5532 | 56.6418 |
| CSCS | 0.0 | 5 | 1 | 18.6724 | 30.9726 | 36.4207 | 37.3043* | 47.6354 | 52.3970 |
| CSCS | 0.0 | 5 | 2 | 18.2212 | 29.6961 | 35.2362 | 36.3293* | 45.8708 | 49.8450 |
| CSCS | 0.0 | 5 | 5 | 17.7219 | 28.2744 | 33.9063 | 35.1538* | 43.8898 | 47.1330 |
| CSCS | 0.1 | 1 | 0 | 22.4465 | 38.6817 | 46.6429 | 53.5088* | 61.5218 | 67.2651 |
| CSCS | 0.1 | 1 | 1 | 19.5096 | 32.6522 | 38.5730 | 41.7582* | 50.7891 | 55.4963 |
| CSCS | 0.1 | 1 | 2 | 18.5650 | 29.7765 | 35.7326 | 38.8950* | 46.4711 | 49.2450 |
| CSCS | 0.1 | 1 | 5 | 17.5729 | 26.8278 | 32.7906 | 35.2034* | 41.9238 | 43.4749 |
| CSCS | 0.1 | 5 | 0 | 19.0452 | 31.2291 | 36.8058 | 39.9613* | 47.9353 | 52.5791 |
| CSCS | 0.1 | 5 | 1 | 17.8831 | 28.8803 | 33.6948 | 34.4487* | 43.8465 | 48.2218 |
| CSCS | 0.1 | 5 | 2 | 17.4194 | 27.5091 | 32.4045 | 33.1878* | 41.9061 | 45.3921 |
| CSCS | 0.1 | 5 | 5 | 16.8926 | 25.9209 | 30.9100 | 31.6721* | 39.6569 | 42.2959 |
| CCCC | 0.0 | 1 | 0 | 26.9220 | 49.6013 | 49.6013 | 69.2348 | 81.6650 | 82.4035 |
| CCCC | 0.0 | 1 | 1 | 23.6411 | 42.3454 | 42.6280 | 59.0121 | 68.9714 | 69.4776 |
| CCCC | 0.0 | 1 | 2 | 22.3022 | 39.0488 | 39.7465 | 54.7868 | 62.4589 | 64.6548 |
| CCCC | 0.0 | 1 | 5 | 20.8932 | 35.5030 | 37.0162 | 50.2457 | 56.5494 | 59.9044 |
| CCCC | 0.0 | 5 | 0 | 23.2387 | 41.1465 | 41.1465 | 56.5383 | 66.0695 | 66.7244 |
| CCCC | 0.0 | 5 | 1 | 21.7421 | 38.0297 | 38.1921 | 52.2836 | 61.0631 | 61.6386 |
| CCCC | 0.0 | 5 | 2 | 21.0438 | 36.4884 | 36.7607 | 50.3269 | 58.3569 | 59.2582 |
| CCCC | 0.0 | 5 | 5 | 20.2676 | 34.6341 | 35.3354 | 48.1178 | 55.3775 | 56.8868 |
| CCCC | 0.1 | 1 | 0 | 26.6373 | 48.8590 | 48.8590 | 68.1777 | 80.4481 | 81.1669 |
| CCCC | 0.1 | 1 | 1 | 22.8924 | 40.4052 | 40.7657 | 56.2899 | 65.5238 | 65.9564 |
| CCCC | 0.1 | 1 | 2 | 21.3380 | 36.5152 | 37.3084 | 51.2542 | 57.6091 | 60.2741 |
| CCCC | 0.1 | 1 | 5 | 19.7169 | 32.3681 | 34.0803 | 45.8241 | 50.7494 | 54.6162 |
| CCCC | 0.1 | 5 | 0 | 22.0711 | 38.4035 | 38.4035 | 52.5487 | 61.3362 | 61.9445 |
| CCCC | 0.1 | 5 | 1 | 20.5961 | 35.1490 | 35.3718 | 48.0627 | 55.9862 | 56.5228 |
| CCCC | 0.1 | 5 | 2 | 19.8603 | 33.4969 | 33.7614 | 45.8926 | 52.9199 | 53.8563 |
| CCCC | 0.1 | 5 | 5 | 19.0136 | 31.3879 | 32.1552 | 43.3653 | 49.4759 | 51.1602 |

Note: Edge conditions are given in the order $x = 0$, $y = 0$, $x = a$, $y = b$, with S = simply supported, C = clamped; 3D = three-dimensional; $a/h$ = length-to-thickness ratio; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; $\alpha_p$ = porosity volume fraction; BC = boundary conditions; $\Omega_1$–$\Omega_6$ = the first six dimensionless frequencies; SSSS = all four edges simply supported; CSCS = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ simply supported; CCCC = all four edges clamped; "*" = marks in-plane modes. Values for plates with clamped edges are upper bounds accurate to about three significant digits.

In-plane grading also makes the modes asymmetric. The peak of the first mode moves toward the metal-rich edge $x = 0$, reaching $x_{\max}/a = 0.41$–$0.46$ at $k_x \approx 2$–$2.5$ depending on $k_z$, and then moves back toward the centre as $k_x$ increases further (Figure 2c). A mode concentrates where the local ratio of stiffness to mass, and hence the local frequency, is lowest; this is the metal-rich part. For large $k_x$ only a narrow strip near $x = a$ is ceramic-rich, the plate is nearly uniform again, and the asymmetry decreases. The shift is largest for small $k_z$, where the top layer is most ceramic-rich and the contrast along the plate is strongest. The foundation reduces the shift slightly (for $k_z = 1$, $x_{\max}/a = 0.435$ on the foundation versus 0.428 without it at $k_x = 2$), because its stiffness raises the local frequency more where the plate is lighter. The 3D results (symbols) confirm the HSDT trends.

**Mode shapes.** Figure 3 shows the first four flexural modes from the 3D model for $k_x = 0, 2$ and 5. For the thickness-graded plate the second and third modes are degenerate (35.831), as for any square plate with symmetric properties. In-plane grading breaks the symmetry between $x$ and $y$ and splits the pair: for $k_x = 2$, the mode with a nodal line parallel to the $x$-axis has $\Omega = 27.725$, and the mode with a nodal line parallel to the $y$-axis has

$\Omega = 27.972$. The nodal lines parallel to the $y$-axis move toward the ceramic-rich edge (to $x/a \approx 0.45$ in the third mode for $k_x = 2$), and the larger antinode lies in the metal-rich half. The same pattern appears in the fourth mode. A plate theory solved by a double Fourier series assumes symmetric mode shapes and cannot represent this distortion.

**Table 7.** 3D benchmark frequencies; $a/h = 10$, SSSS, no foundation

| BC | $\alpha_p$ | $k_z$ | $k_x$ | $\Omega_1$ | $\Omega_2$ | $\Omega_3$ | $\Omega_4$ | $\Omega_5$ | $\Omega_6$ |
|---|---|---|---|---|---|---|---|---|---|
| SSSS | 0.0 | 1 | 0 | 14.6196 | 35.1022 | 35.1022 | 53.5156* | 53.5156* | 54.1593 |
| SSSS | 0.0 | 1 | 1 | 12.5884 | 30.0164 | 30.0570 | 43.6125* | 44.4196* | 46.2690 |
| SSSS | 0.0 | 1 | 2 | 11.6426 | 27.5054 | 27.7460 | 40.6707* | 41.3203* | 42.8249 |
| SSSS | 0.0 | 1 | 5 | 10.6503 | 24.9349 | 25.4084 | 36.5498* | 38.3674* | 39.1999 |
| SSSS | 0.0 | 5 | 0 | 12.4543 | 29.4795 | 29.4795 | 41.7303* | 41.7303* | 44.9704 |
| SSSS | 0.0 | 5 | 1 | 11.4020 | 27.0139 | 27.0651 | 37.3043* | 37.4850* | 41.3039 |
| SSSS | 0.0 | 5 | 2 | 10.8521 | 25.6944 | 25.7873 | 35.9252* | 36.3293* | 39.5386 |
| SSSS | 0.0 | 5 | 5 | 10.2588 | 24.2207 | 24.4775 | 34.3034* | 35.1538* | 37.6111 |
| SSSS | 0.1 | 1 | 0 | 14.3071 | 34.3925 | 34.3925 | 53.1180 | 53.5088* | 53.5088* |
| SSSS | 0.1 | 1 | 1 | 11.9591 | 28.4807 | 28.5191 | 41.7582* | 42.9269* | 43.9600 |
| SSSS | 0.1 | 1 | 2 | 10.8262 | 25.4485 | 25.7617 | 38.4077* | 38.8950* | 39.8397 |
| SSSS | 0.1 | 1 | 5 | 9.6393 | 22.3907 | 22.9772 | 33.3531* | 35.2034* | 35.4932 |
| SSSS | 0.1 | 5 | 0 | 11.5117 | 27.2355 | 27.2355 | 39.9613* | 39.9613* | 41.5402 |
| SSSS | 0.1 | 5 | 1 | 10.4511 | 24.7095 | 24.7802 | 34.4487* | 34.7421* | 37.7498 |
| SSSS | 0.1 | 5 | 2 | 9.8399 | 23.2513 | 23.3387 | 32.7775* | 33.1878* | 35.7789 |
| SSSS | 0.1 | 5 | 5 | 9.1634 | 21.5561 | 21.8533 | 30.7184* | 31.6721* | 33.5775 |

Note: 3D = three-dimensional; $a/h$ = length-to-thickness ratio; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; $\alpha_p$ = porosity volume fraction; BC = boundary conditions; $\Omega_1$–$\Omega_6$ = the first six dimensionless frequencies; SSSS = all four edges simply supported; "*" = marks in-plane modes.

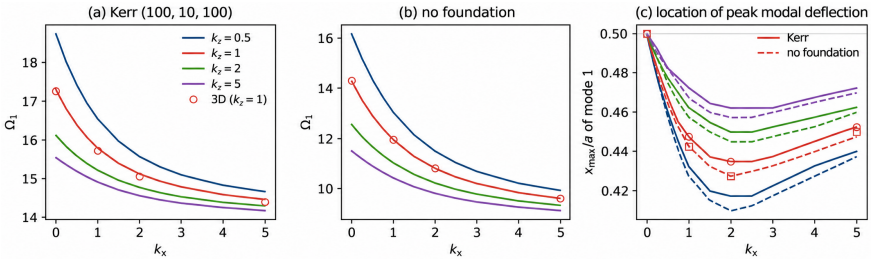



**Figure 2.** Effect of the gradation indices on: (a) the fundamental frequency on the Kerr foundation; (b) the fundamental frequency without a foundation; (c) the location of the peak of the first mode along $y = b/2$; SSSS, $a/h = 10$, $\alpha_p = 0.1$, Lines: HSDT, symbols: 3D

Note: SSSS = all four edges simply supported; $a/h$ = length-to-thickness ratio; $\alpha_p$ = porosity volume fraction; $k_x$ = gradation index along the length; HSDT = higher-order shear deformation theory; 3D = three-dimensional.

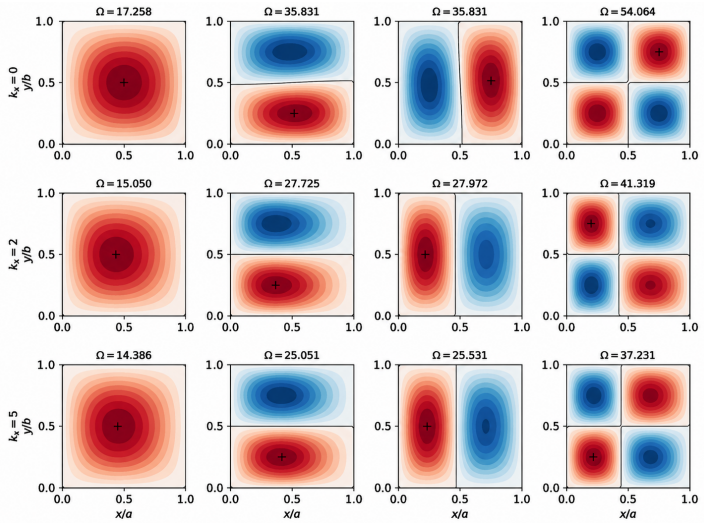



**Figure 3.** First four flexural mode shapes (mid-plane deflection, 3D model) of plates on the Kerr foundation (100, 10, 100) for $k_x = 0$ (top), 2 (middle) and 5 (bottom); SSSS, $a/h = 10$, $k_z = 1$, $\alpha_p = 0.1$

Note: SSSS = all four edges simply supported; $a/h$ = length-to-thickness ratio; $\alpha_p$ = porosity volume fraction; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; 3D = three-dimensional; black lines = nodal lines; crosses = peak amplitude.

**Kerr foundation.** Figure 4 shows the effect of the three Kerr parameters on the fundamental flexural frequency and on the location of its peak. As $K_u$ increases, the frequency rises from its value without a foundation (10.82) to the Pasternak limit with the same $K_s$ and $K_l$ (dotted lines). For $K_s = 0$ it rises to the Winkler limit with modulus $K_l$ (16.30). The upper spring thus acts in series with the two lower layers, as in static problems [30]. For $K_u \lesssim 30$ the upper spring governs and the other two parameters hardly matter. For $K_u \gtrsim 300$ the shear layer and the lower spring control the response and can raise the frequency by a factor of up to five. The 3D results (stars) confirm the HSDT trend: 11.46, 15.05, 21.25 and 23.20 at $K_u$ = 10, $10^2$, $10^3$, and $10^4$, versus 11.47, 15.12, 21.43, and 23.41 from the HSDT.

A stiff foundation has two further effects that are specific to bidirectional plates. First, it moves the peak of the first mode from the metal-rich side across the centre. For $(K_s, K_l) = (100, 100)$ the peak reaches $x_{\max}/a = 0.51$ at large $K_u$, on the ceramic-rich side. When the foundation dominates the stiffness, the local frequency scales with $\sqrt{k_{\text{eq}}/(\rho h)}$ and is lowest where the plate is heaviest, i.e. near the ceramic-rich edge. Second, the foundation raises only the flexural frequencies. The first in-plane mode ($\Omega \approx 38.4$, chain line) is practically unaffected, so for stiff foundations it becomes the fundamental mode of the plate. This occurs for $K_u$ between about $1.6 \times 10^3$ and $2.5 \times 10^3$ when $(K_s, K_l) = (100, 100)$, and between $2.5 \times 10^3$ and $4 \times 10^3$ when $(K_s, K_l) = (10, 1000)$. Near the crossing the in-plane frequency changes slightly (between 38.1 and 38.6), because in-plane grading couples bending and stretching. Analyses that track only the transverse displacement, or that assume the lowest mode to be flexural, miss this change of the fundamental mode.

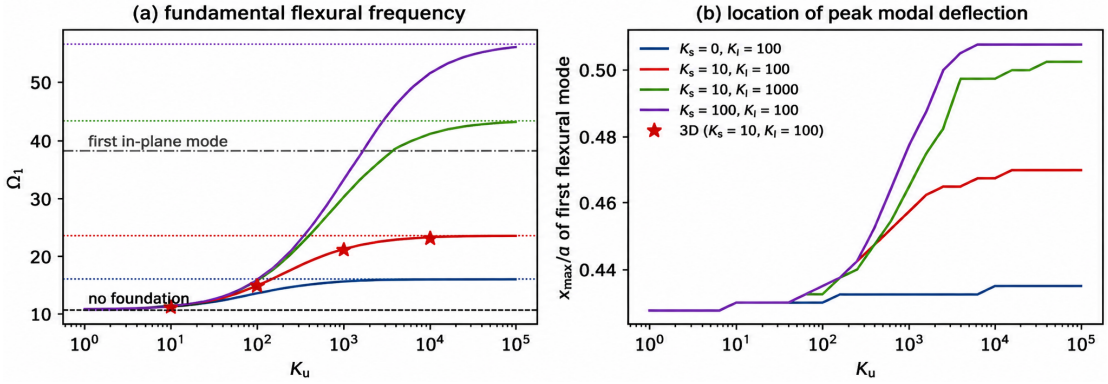



**Figure 4.** Effect of the Kerr parameters on: (a) the fundamental flexural frequency; (b) the location of its peak along $y = b/2$; HSDT, SSSS, $a/h = 10$, $k_x = 2$, $k_z = 1$, $\alpha_p = 0.1$

Note: SSSS = all four edges simply supported; HSDT = higher-order shear deformation theory; $a/h$ = length-to-thickness ratio; $\alpha_p$ = porosity volume fraction; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; dotted lines = Pasternak/Winkler limits ($K_u \to \infty$); dashed line = no foundation; chain line = first in-plane mode; stars = three-dimensional (3D) results.

**Porosity.** Porosity lowers both the modulus and the density, and Table 8 and Figure 5 show that its net effect depends on the foundation. Without a foundation the frequency decreases with $\alpha_p$, because Eq. (2) removes a larger fraction of the stiffness than of the mass, particularly in the metal-rich regions. Raising $\alpha_p$ from 0 to 0.2 lowers the 3D frequency only slightly for $k_x = 0$, but considerably for $k_x = 2$ and even more for $k_x = 5$, since bidirectional plates are mostly metallic. On the reference Kerr foundation, part of the stiffness is supplied by the foundation and is not affected by porosity. The decrease is then small for the bidirectional plates, and the frequency of the thickness-graded plate is practically independent of $\alpha_p$ (17.25–17.26 for $\alpha_p \leq 0.15$). On the stiff foundation (1000, 100, 1000) the trend reverses. The foundation dominates the stiffness, porosity mainly reduces the mass, and the fundamental flexural frequency increases appreciably with $\alpha_p$ for both $k_x = 0$ and $k_x = 2$. The in-plane frequencies, which the foundation does not raise, decrease with porosity. As a result, for $k_x = 5$ with $\alpha_p \geq 0.1$, and for $k_x = 2$ with $\alpha_p = 0.2$, an in-plane mode becomes the fundamental mode (marked † in Table 8). Porosity therefore cannot be said to lower the natural frequencies in general: the sign of its effect is set by the ratio of foundation to plate stiffness. The HSDT reproduces all these trends; its differences from the 3D results are negligible without a foundation, small on the reference foundation and somewhat larger on the stiff foundation.

**Edge conditions.** Table 9 lists HSDT results for seven edge combinations. As expected, clamping raises the frequencies, and the CCCC plate has the highest fundamental frequency (26.77, 21.52 and 19.86 for $k_x = 0, 2$ and 5). For the thickness-graded plate the CSCS and SCSC plates are identical by symmetry. In a bidirectional plate they differ: clamping the graded edges $x = 0, a$ (CSCS) gives a higher fundamental frequency (18.81 for $k_x = 2$) than clamping the ungraded edges $y = 0, b$ (SCSC, 18.65), but a lower second frequency. SCSC also gives the largest shift of the peak toward the metal-rich edge ($x_{\max}/a = 0.408$ for $k_x = 2$), because the simply supported graded edges leave the soft metal-rich region free to rotate. The CFCF plate has closely spaced first and second frequencies

(15.64 and 17.87 for $k_x = 2$). In all cases the peak of the first mode lies in the metal-rich half of the plate.

**Table 8.** Effect of porosity on the fundamental flexural frequency; SSSS, $a/h = 10$, $k_z = 1$

| $k_x$ | $\alpha_p$ | No fdn: 3D | HSDT | (100, 10, 100): 3D | HSDT | (1000, 100, 1000): 3D | HSDT |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 14.6196 | 14.5984 | 17.2541 | 17.2950 | 32.3795 | 32.7578 |
| 0 | 0.05 | 14.4766 | 14.4549 | 17.2596 | 17.3050 | 32.9685 | 33.3767 |
| 0 | 0.1 | 14.3071 | 14.2848 | 17.2579 | 17.3085 | 33.6029 | 34.0469 |
| 0 | 0.15 | 14.1037 | 14.0807 | 17.2462 | 17.3029 | 34.2871 | 34.7755 |
| 0 | 0.2 | 13.8567 | 13.8326 | 17.2206 | 17.2846 | 35.0245 | 35.5708 |
| 2 | 0 | 11.6426 | 11.6362 | 15.2405 | 15.2950 | 33.0444 | 33.4304 |
| 2 | 0.05 | 11.2733 | 11.2688 | 15.1560 | 15.2178 | 33.7582 | 34.1832 |
| 2 | 0.1 | 10.8262 | 10.8242 | 15.0498 | 15.1208 | 34.5321 | 35.0029 |
| 2 | 0.15 | 10.2688 | 10.2703 | 14.9124 | 14.9952 | 35.3674 | 35.8472 |
| 2 | 0.2 | 9.5421 | 9.5489 | 14.7256 | 14.8252 | 36.2631† | 37.0209† |
| 5 | 0 | 10.6503 | 10.6445 | 14.6266 | 14.6791 | 33.2709 | 33.6410 |
| 5 | 0.05 | 10.1937 | 10.1894 | 14.5174 | 14.5765 | 33.9956 | 34.3190 |
| 5 | 0.1 | 9.6393 | 9.6368 | 14.3860 | 14.4532 | 34.8975† | 35.4499† |
| 5 | 0.15 | 8.9453 | 8.9449 | 14.2235 | 14.3010 | 35.7568† | 36.4021† |
| 5 | 0.2 | 8.0343 | 8.0365 | 14.0141 | 14.1060 | 36.6638† | 37.4957† |

Note: SSSS = all four edges simply supported; HSDT = higher-order shear deformation theory; $a/h$ = length-to-thickness ratio; $\alpha_p$ = porosity volume fraction; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; 3D = three-dimensional; No fdn = a plate without a foundation; (100, 10, 100) and (1000, 100, 1000) = the Kerr parameters $(K_u, K_s, K_l)$; † = cases in which an in-plane mode has a lower frequency than the first flexural mode.

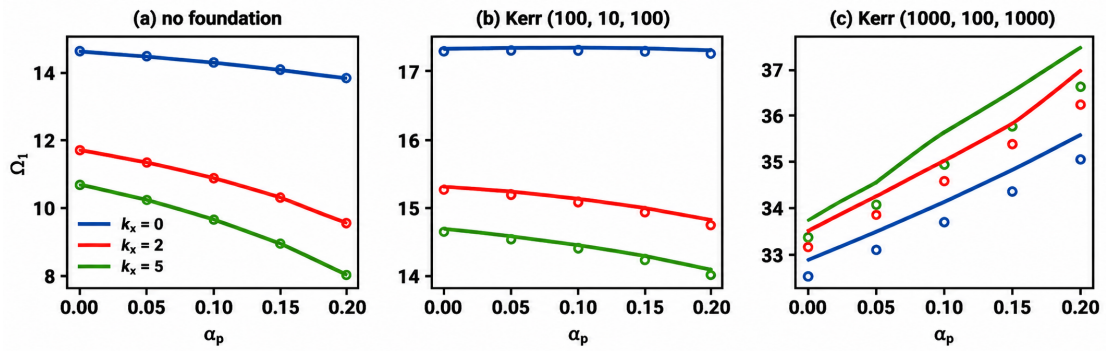



**Figure 5.** Effect of porosity on the fundamental flexural frequency: (a) without a foundation; (b) on the Kerr foundation (100, 10, 100); (c) on the stiff Kerr foundation (1000, 100, 1000); SSSS, $a/h = 10$, $k_z = 1$

Note: SSSS = all four edges simply supported; $a/h$ = length-to-thickness ratio; $\alpha_p$ = porosity volume fraction; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; (100, 10, 100) and (1000, 100, 1000) = the Kerr parameters $(K_u, K_s, K_l)$.

**Table 9.** Effect of the edge conditions; HSDT, $a/h = 10$, $k_z = 1$, $\alpha_p = 0.1$, Kerr (100, 10, 100)

| BC | $k_x = 0$ | | | $k_x = 2$ | | | $k_x = 5$ | | |
|---|---|---|---|---|---|---|---|---|---|
| | $\Omega_1$ | $\Omega_2$ | $x_{\max}/a$ | $\Omega_1$ | $\Omega_2$ | $x_{\max}/a$ | $\Omega_1$ | $\Omega_2$ | $x_{\max}/a$ |
| SSSS | 17.3085 | 35.7975 | 0.500 | 15.1208 | 27.7648 | 0.435 | 14.4532 | 25.0952 | 0.453 |
| SSSF | 12.7315 | 22.2917 | 0.500 | 12.0462 | 18.6248 | 0.455 | 11.8534 | 17.5149 | 0.468 |
| CSCS | 22.6937 | 39.0141 | 0.500 | 18.8143 | 30.1525 | 0.440 | 17.7554 | 27.1018 | 0.453 |
| SCSC | 22.6937 | 39.0141 | 0.500 | 18.6516 | 30.3677 | 0.408 | 17.2718 | 27.6303 | 0.435 |
| CSCF | 19.0587 | 27.0377 | 0.500 | 16.2823 | 21.9272 | 0.445 | 15.6677 | 20.4027 | 0.455 |
| CFCF | 18.1821 | 21.1933 | 0.500 | 15.6370 | 17.8654 | 0.448 | 15.1094 | 17.0557 | 0.455 |
| CCCC | 26.7696 | 49.5027 | 0.500 | 21.5226 | 37.1807 | 0.430 | 19.8571 | 32.9194 | 0.448 |

Note: Edge conditions are given in the order $x = 0$, $y = 0$, $x = a$, $y = b$, with S = simply supported, C = clamped, and F = free; HSDT = higher-order shear deformation theory; $a/h$ = length-to-thickness ratio; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; $\alpha_p$ = porosity volume fraction; $x_{\max}$ = location of the peak of the first mode along $y = b/2$; BC = boundary conditions; $\Omega_1$ and $\Omega_2$ = the first two flexural dimensionless frequencies; SSSS = all four edges simply supported; SSSF = edge $y = b$ free, other edges simply supported; CSCS = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ simply supported; SCSC = edges $y = 0$ and $y = b$ clamped, edges $x = 0$ and $x = a$ simply supported; CSCF = edges $x = 0$ and $x = a$ clamped, edge $y = 0$ simply supported, edge $y = b$ free; CFCF = edges $x = 0$ and $x = a$ clamped, edges $y = 0$ and $y = b$ free; CCCC = all edges clamped.

**Thickness ratio.** Table 10 compares the three models over a wide range of thickness ratios. The HSDT converges to the 3D solution as the plate becomes thin, and the two practically coincide at $a/h = 50$ for both $k_x = 0$ and

$k_x = 5$. The quasi-3D theory converges to a value that remains noticeably too high for $k_x = 0$. For thick plates ($a/h = 4$) both theories clearly overestimate the frequency.

**Table 10.** Fundamental frequency versus thickness ratio; SSSS, $k_z = 1$, $\alpha_p = 0.1$, Kerr (100, 10, 100)

| $k_x$ | $a/h$ | 3D | HSDT | Quasi-3D |
|---|---|---|---|---|
| 0 | 4 | 15.4143 | 15.7630 | 15.8446 |
| 0 | 5 | 16.1150 | 16.3288 | 16.4653 |
| 0 | 10 | 17.2579 | 17.3085 | 17.5235 |
| 0 | 20 | 17.5986 | 17.6111 | 17.8472 |
| 0 | 50 | 17.6992 | 17.7012 | 17.9433 |
| 5 | 4 | 12.8248 | 13.0941 | 12.9513 |
| 5 | 5 | 13.4799 | 13.7791 | 13.6122 |
| 5 | 10 | 14.3860 | 14.4532 | 14.4258 |
| 5 | 20 | 14.6483 | 14.6648 | 14.6721 |
| 5 | 50 | 14.7259 | 14.7285 | 14.7457 |

Note: 3D = three-dimensional; SSSS = all four edges simply supported; $a/h$ = length-to-thickness ratio; $k_x$ = gradation index along the length; $k_z$ = gradation index through the thickness; $\alpha_p$ = porosity volume fraction; HSDT = higher-order shear deformation; Quasi-3D = reduced-unknown quasi-3D theory theory.

## 5 Conclusions

The free vibration of length- and thickness-graded porous plates resting on a three-parameter Kerr foundation was studied with a 3D elasticity Ritz model and two plate theories. The main conclusions are:

1. A 3D Ritz model with a statically condensed Kerr foundation reproduces an independent exact state-space solution to all reported digits and the classical results of Srinivas et al. for homogeneous plates. New 3D benchmark frequencies are provided for bidirectional porous plates with and without a Kerr foundation (Tables 6 and 7).
2. In-plane grading lowers the natural frequencies, because aluminium has a much lower stiffness-to-density ratio than alumina. From $k_x = 0$ to $k_x = 5$ the fundamental frequency falls by about one third without a foundation, and moderately on the reference Kerr foundation.
3. In-plane grading makes the mode shapes asymmetric. The peak of the first mode moves toward the metal-rich edge (to $x/a \approx 0.41$–$0.46$ for $k_x \approx 2$), degenerate mode pairs of square plates split, and nodal lines move toward the ceramic-rich edge. A stiff foundation moves the peak back across the centre toward the heavier, ceramic-rich edge.
4. The HSDT agrees closely with the 3D solution for SSSS plates with $a/h \geq 10$, and its error vanishes for thin plates. The reduced-unknown quasi-3D theory is not more accurate. For thickness-graded plates it retains a noticeable error even for $a/h = 50$, a consequence of thickness locking. This error decreases with in-plane grading.
5. The Kerr upper spring acts in series with the shear layer and the lower spring. Stiff foundations raise only the flexural frequencies, so the first in-plane mode, which is coupled to bending in bidirectional plates, can become the fundamental mode of the plate.
6. The effect of porosity depends on the foundation. It lowers the frequency of an unsupported plate, considerably so for strongly graded bidirectional plates, has little effect on the reference foundation and raises the flexural frequency on a stiff foundation.
7. In bidirectional plates, clamping the graded edges and clamping the ungraded edges give different frequencies and mode shapes, while the two cases coincide for thickness-graded plates.

For sensitive structural designs involving thick plates, clamped edges or stiff foundations, simplified 2D and reduced-unknown quasi-3D models should be checked against exact 3D elasticity or benchmarked Ritz solutions, since thickness locking and the coupling of bending and stretching can affect the predicted frequencies and modes.

Further work can address forced and transient vibration, thermal pre-stress, and the vibration of plates on tensionless foundations, for which the contact region must be determined as part of the solution.

### Data Availability

The Python code used to generate all results and figures is available from the author on reasonable request.

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**Conflicts of Interest**

The author declares no conflicts of interest.

**Declaration on the Use of Generative AI and AI-assisted Technologies**

Generative AI tools (Gemini) were used exclusively for grammatical checking, stylistic refinement, and linguistic phrasing.

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